

Tractable Problems in AI Security via Formal Methods

A sane way to burn a few postdoc-years

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1 The strategic landscape

Let us treat ML training and inference with the seriousness of airplanes



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We mostly write about model weight confidentiality and integrity, but other notions of AI security are in scope. Not the biggest possible tent, but more than two invariants.



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Figure 1: Demands hardened critical infrastructure, doesn't say how.

Verified Machine Learning Infrastructure

Formal Methods for Trustworthy Artificial Intelligence Deployment

[Gopal P. Sarma](#), [Rachel Steratore](#), [Sunny D. Bhatt](#), [Geoffrey Irving](#)

RESEARCH — Published Jun 4, 2026



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ADDITIONAL DOWNLOADS

Includes Annex: Survey Instrument

📄 One-page overview

Figure 2: Expert survey, human and simulated/silicon

ARIA

TODO



Anthropic's Frontier Safety Roadmap

We believe that AI capabilities will improve rapidly in the coming years. We will need to quickly and dramatically improve our state of preparedness in a number of areas, especially:

Security

Preventing theft, sabotage and/or manipulation of our AI models.

Safeguards

Preventing dangerous use of our models via product surfaces and within Anthropic itself.

Figure 3: See *Leveling up across the board* at <https://www.anthropic.com/responsible-scaling-policy/roadmap>

Speed

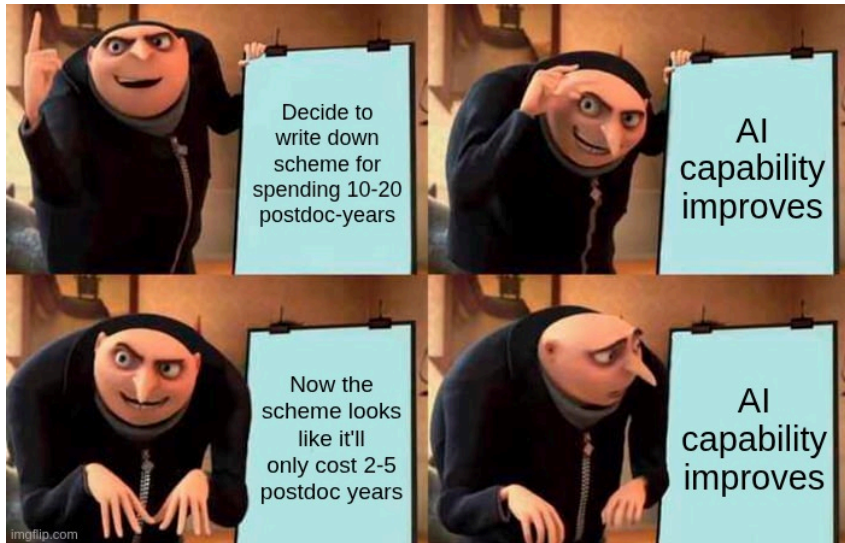


Figure 4: This is technically a violation of the meme format.

2 The stack

5 layers



Figure 5: The path of a prompt from the enduser to the chip's circuits

3 Tractable Problems

Hardening protocol boundaries



TODO

Hardening OCI runtime



TODO

GPU drivers for verified kernels



TODO

4 Onwards

How you can help



More contributions

Come aboard! More room for authors. Or, if you want to be low-key, comment on the website's native comments feature.

Tokens

Ask claude to implement one of the project sketches on `/problems`. Let us know how it goes.

Podcasts

What podcasts should we go on to discuss this? Get us invited.

I'm easy to find

`quinn@for-all.dev`